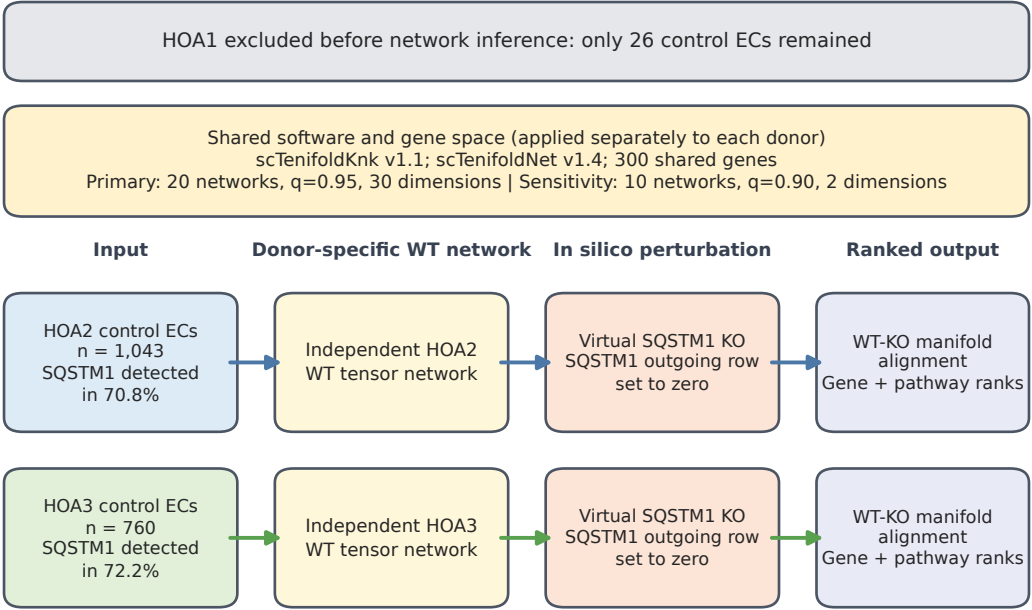
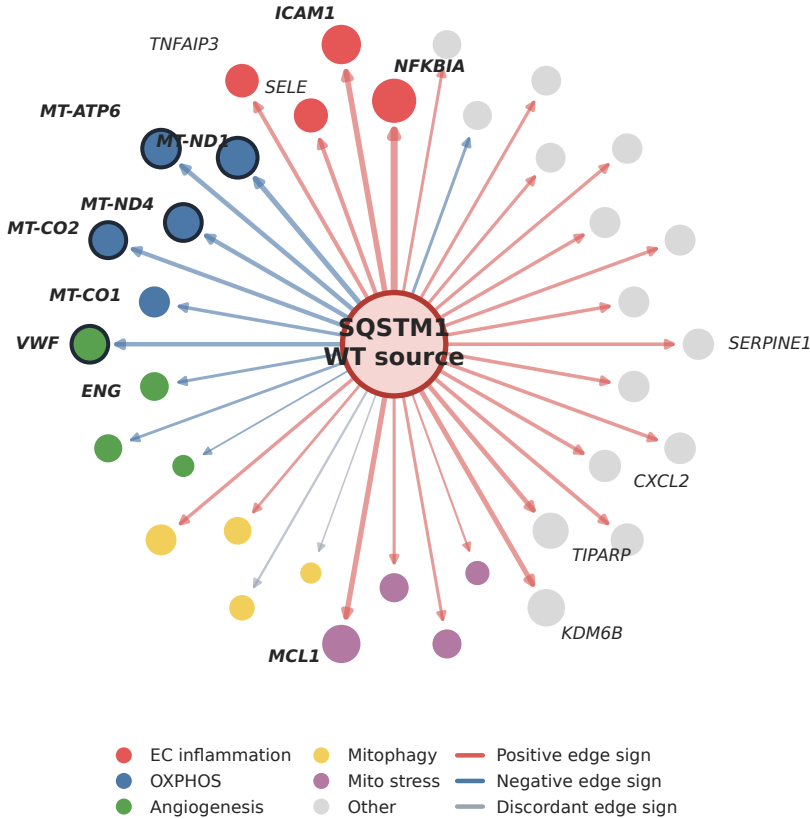


A Donor-separated official-package virtual-knockout workflow

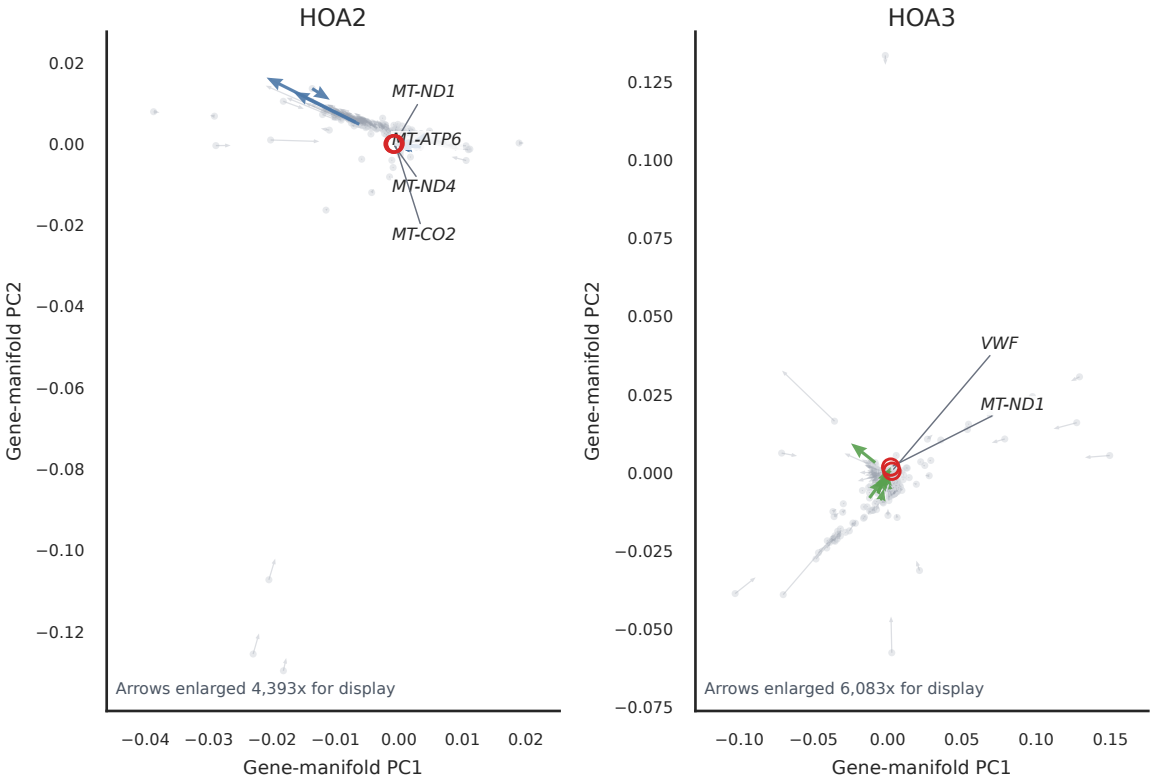


Guardrail: HOA2 and HOA3 used separate expression matrices, tensor networks, and WT-KO alignments; donor data were not pooled.

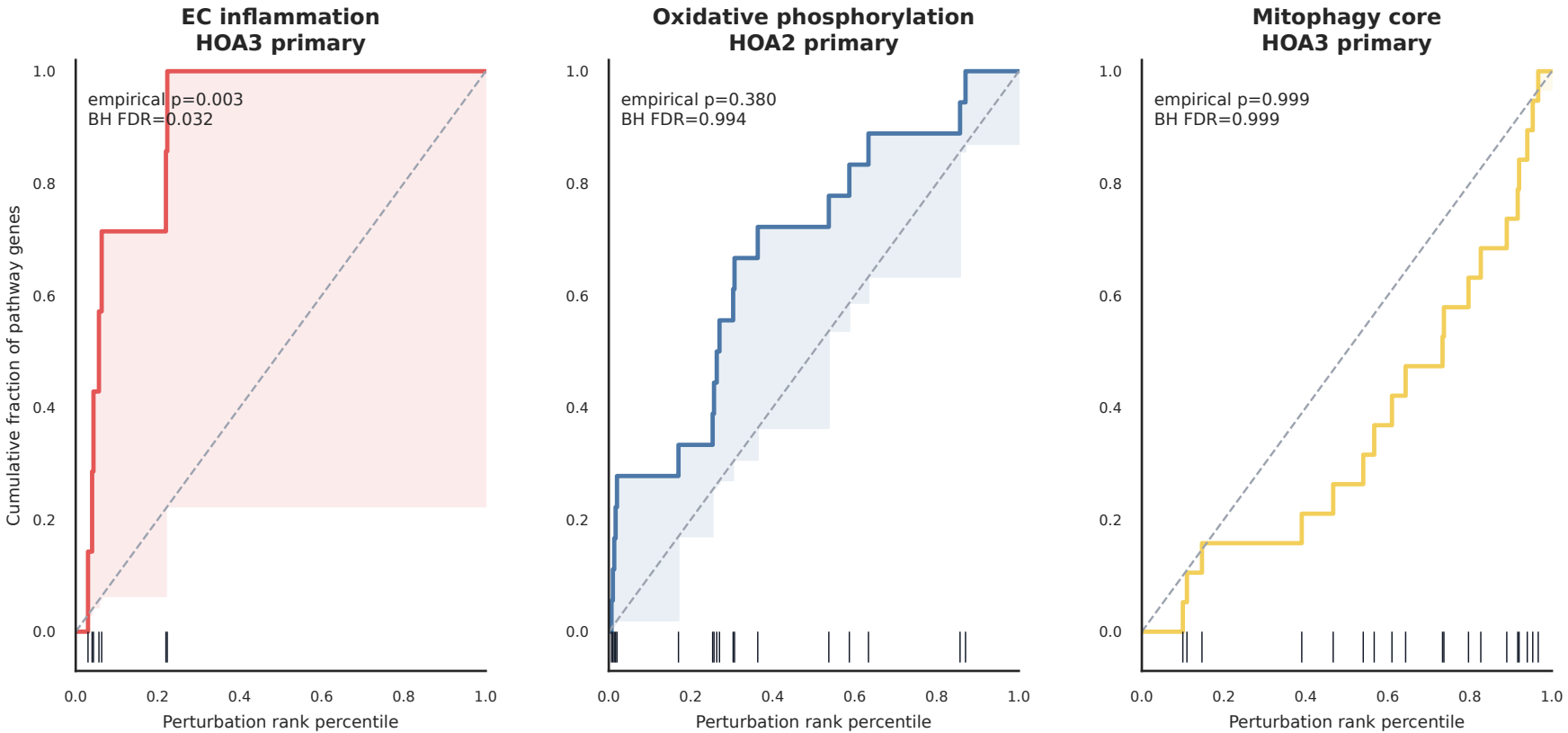
B SQSTM1 WT neighborhood removed by virtual KO



C WT-to-virtual-KO displacement in the aligned gene manifold



D Matched-null pathway rank recovery



Gray arrows: downstream genes; colored arrows: top 10 ranks; red rings/labels: FDR<0.05.
Uniform magnification preserves direction/relative size; these are gene-manifold displacements, not cell-fate vectors.

Curves show pathway-member recovery along official perturbation ranks; p/FDR use 20,000 matched null sets.
These descriptive curves are not GSEA enrichment-score curves.